2.2 Classwork: Radicals and Exponents (no calculator) — Markscheme

1. [Maximum mark: 14]

(a) $3^4 \times 3^2 = 3^{4+2} = 3^6$ **A1**

Accept 729.

(b) $\frac{x^7}{x^3} = x^{7-3} = x^4$ **A1**

(c) $(a^2)^3 = a^{2\times 3} = a^6$ **A1**

(d) $(2x^3)^2 = 2^2 x^{3\times 2} = 4x^6$ **A1A1**

A1 for the coefficient 4, *A1 for* x^6. *The common error is* $2x^6$*: A0A1.*

(e) $\frac{6x^5}{2x^2} = \frac{6}{2}x^{5-2} = 3x^3$ **A1A1**

A1 for the coefficient 3, *A1 for* x^3.

(f) $7^0 = 1$ **A1**

(g) $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$ **A1**

Accept 0.125. *The common error is* -8 *or* -6*: A0.*

(h) $9^{\frac{1}{2}} = \sqrt{9} = 3$ **A1**

(i) $8^{\frac{2}{3}} = \left(\sqrt[3]{8}\right)^2 = 2^2$ **M1**

$= 4$ **A1**

Accept $\sqrt[3]{8^2} = \sqrt[3]{64}$ *for the M1.*

(j) $4^{-\frac{1}{2}} = \frac{1}{4^{\frac{1}{2}}} = \frac{1}{\sqrt{4}}$ **M1**

$= \frac{1}{2}$ **A1**

Accept 0.5. *Each one-mark part may be answered with the value alone. In (i) and (j) a correct value with no working earns A1 only.*

2. [Maximum mark: 15]

(a) $\sqrt{12} = \sqrt{4}\sqrt{3}$ **M1**

$= 2\sqrt{3}$ **A1**

(b) $\sqrt{50} = \sqrt{25}\sqrt{2}$ **M1**

$= 5\sqrt{2}$ **A1**

In (a) and (b), M1 for writing the number as a perfect square times another factor. A correct answer with no working earns A1 only.

(c) $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$ **A1**

(d) $3\sqrt{5} + 4\sqrt{5} = 7\sqrt{5}$ **A1**

The common error is $7\sqrt{10}$*: A0.*

(e) $\sqrt{18} - \sqrt{2} = 3\sqrt{2} - \sqrt{2}$ **M1**

$= 2\sqrt{2}$ **A1**

The common error is $\sqrt{16} = 4$*: M0A0.*

(f) $\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ **M1**

$= \frac{6\sqrt{3}}{3}$

$= 2\sqrt{3}$ **A1**

Accept $\frac{6\sqrt{3}}{3}$ *for the A1.*

(g) $\left(3 - \sqrt{5}\right)\left(3 + \sqrt{5}\right) = 9 + 3\sqrt{5} - 3\sqrt{5} - 5$ **M1**

$= 4$ **A1**

Accept the difference of two squares, $3^2 - \left(\sqrt{5}\right)^2$*, for the M1.*

(h) $\frac{8}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}}$ **M1**

$= \frac{8\left(3+\sqrt{5}\right)}{4}$ **A1**

$= 2\left(3 + \sqrt{5}\right)$

$= 6 + 2\sqrt{5}$ **A1**

M1 for multiplying numerator and denominator by the conjugate $3 + \sqrt{5}$*. A1 for the denominator* 4*, FT from (g). Accept* $2\left(3 + \sqrt{5}\right)$ *for the final A1.*